Maritime Communication Severity Classification & Information Extraction

SEAALERT

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Review & Refined Project Definition

Maritime radio calls are made under extreme conditions storms, engine noise, and panic causing critical words like 'MAYDAY' to become unrecognizable in speech recognition. Furthermore, no labeled dataset of maritime radio transcriptions exists, making it impossible to train models on real-world data.

INPUT

Noisy ASR transcript of a maritime radio communication

Simulated VHF: Distress | Urgency | Safety Messages

OUTPUT

Classification: One of four severity levels

Distress | Urgency | Safety | Routine

Information Extraction: Structured fields

Location | Vessel Name | Persons on Board | Nature of Incident | Call Sign / MMSI (optional)

RoBERTa-> fine-tuned on synthetic clean text maritime messages (GPT-4 generated)

Review & Refined Project Definition

DATA

- 1,872 GPT-4 synthetic maritime messages (468 per class)
- Clean | ASR text (SNR: 18 | 12 | 6 dB)
- With | without codewords
- Audio (TTS) + Whisper ASR
- Multiple styles & scenarios

MODELS

- Transformers: RoBERTa (selected) | DistilBERT
- Baselines: TF-IDF (BoW) + Logistic Regression | Linear SVM | Naive Bayes
- Infrastructure: GPT-4 | Faster-Whisper | Coqui TTS
- RoBERTa-> fine-tuned on synthetic clean text maritime messages (GPT-4 generated)

METRICS

- F1-score: primary classification metric
- Accuracy: secondary comparison metric

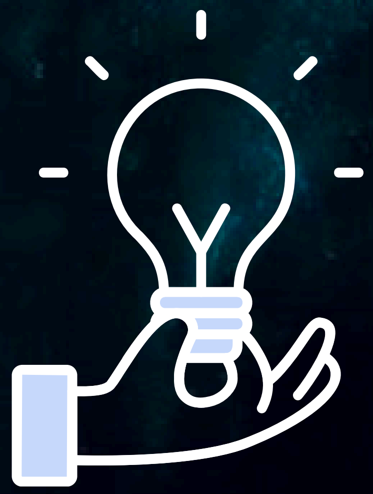
Project achievements and novelty

PROJECT ACHIEVEMENTS

- End-to-end pipeline for maritime message severity classification and information extraction
- Realistic radio-communication pipeline: text → audio → noise → ASR → NLP
- Controlled comparison: Traditional Models vs. Transformer (RoBERTa)

DELIVERABLES

- Severity classification (Distress | Urgency | Safety | Routine)
- Structured operational output (JSON): vessel, location, POB, incident
- Synthetic maritime dataset (text + audio + noisy ASR)

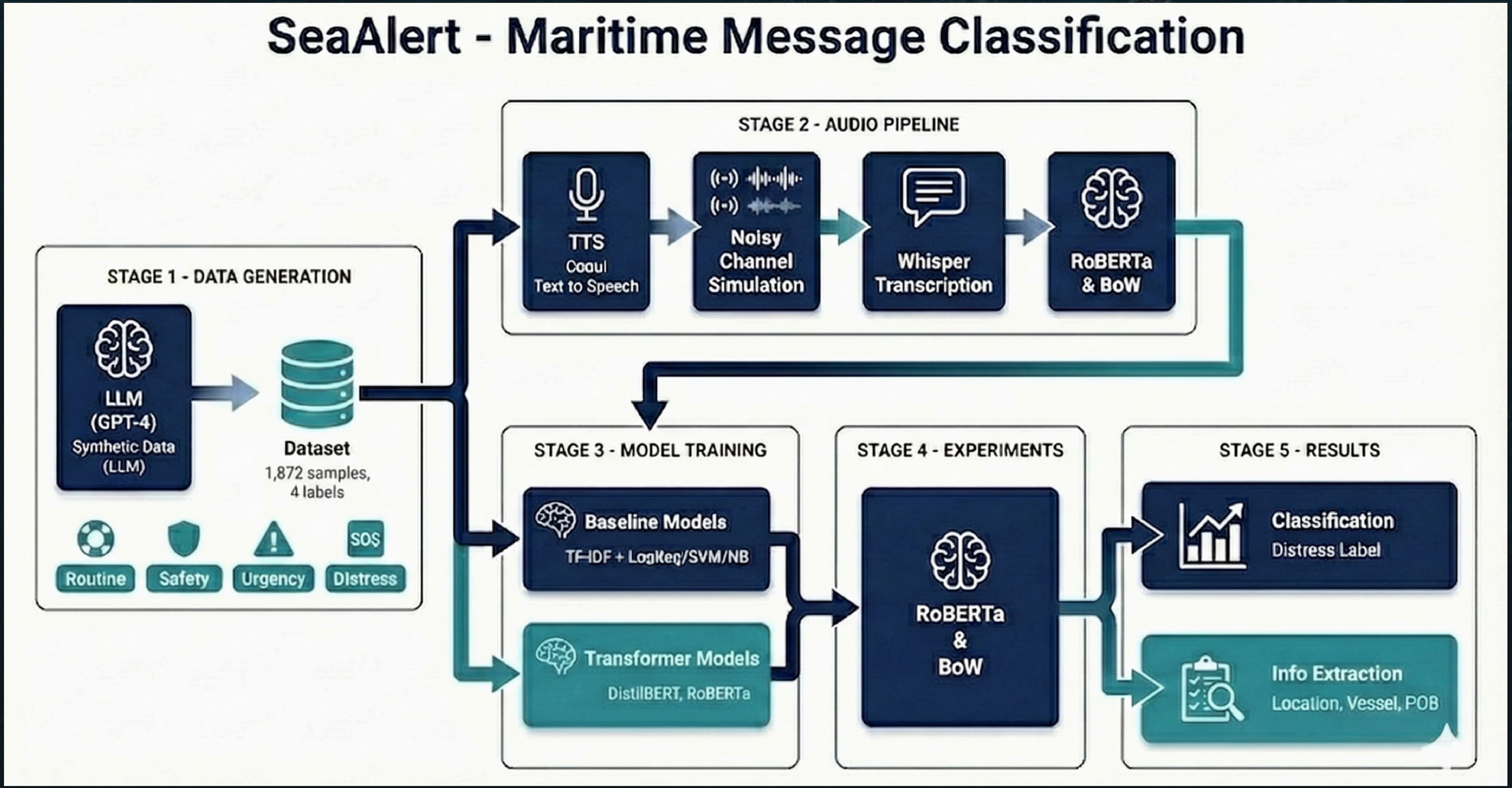


The project's novelty lies in training the system on noisy ASR outputs rather than assuming clean transcripts.

Review Methodology

Dual Augementation

TEXT → TTS → NOISE → ASR → CLASSIFIER → LABEL + EXTRACTION



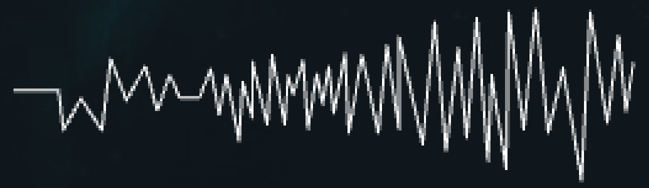
- Synthetic data generation (text + audio)
- Controlled ASR noise simulation (SNR-based)
- Model training & evaluation (F1, Accuracy)
- Robustness & adversarial testing

Example

real example from dataset – demonstrating how severe ASR errors can be under high noise conditions

ORIGINAL MESSAGE

"MAYDAY, MAYDAY, MAYDAY! This is the sailing yacht 'Wind Whisper', uh, we're, um, having a steering failure—can't control the rudder. We're drifting, uh, southeast of, uh, Cape Cod, um, about 10 nautical miles off the shore, I think. Weather's, uh, kinda rough, waves are, like, 3 meters, and it's, um, getting windy. We've got, uh, three people onboard. Need assistance, like, right now! Over."



ASR OUTPUT HIGH NOISE

"maybe, maybe, maybe, this is the sailing yacht with this boat out here, I'm having a steering failure, can't control the rudder, we drifting, I've surfaced through, I kick hard, I'm about 10 knots, I come out as off as sure I think, where those are, can't laugh, waves are like three meters, and it's, I'm getting in the, we've got a few people on board, need the systems right right now, over."

MODEL OUTPUT

Classification

DISTRESS

Extracted Information

- Vessel: NONE
- Location: NONE
- POB: NONE
- Nature: Steering failure, drifting

Critical ASR Errors Shown:

MAYDAY, MAYDAY, MAYDAY → maybe, maybe, maybe (codeword completely lost!)

Wind Whisper → with this boat out here (vessel name destroyed)

Cape Cod → NONE (location corrupted)

nautical miles off the shore → knots, I come out as off as sure (nonsensical)

Results

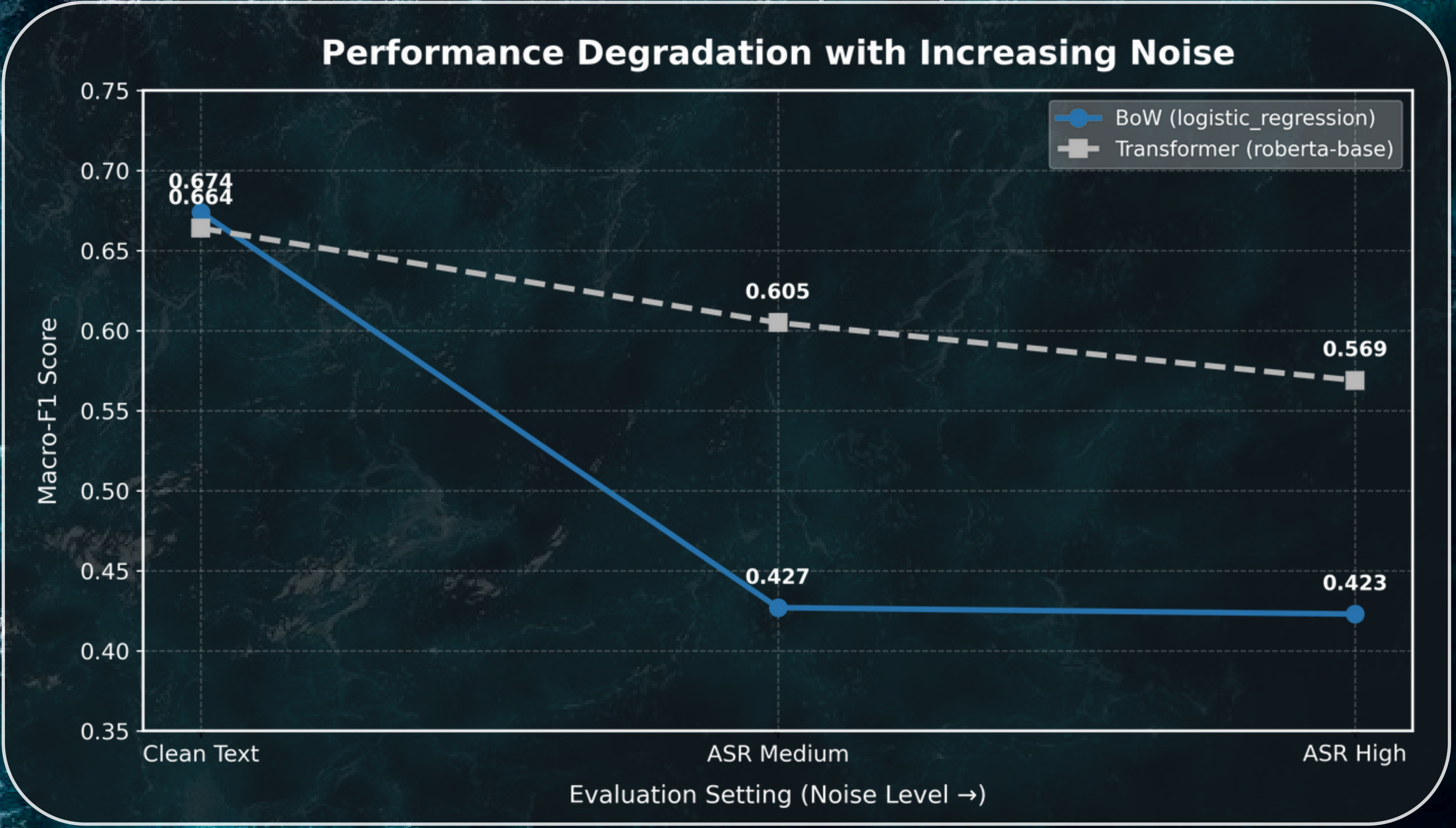
Model Selection:

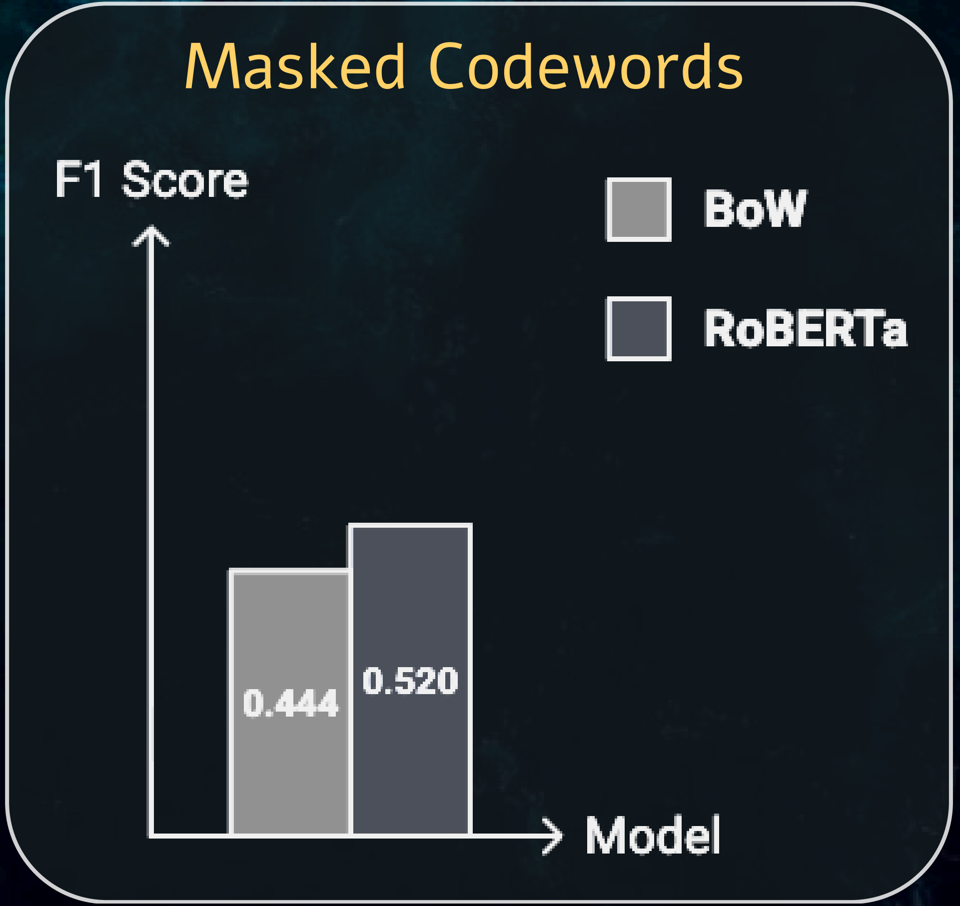
Validation F1 = 0.734 → RoBERTa selected

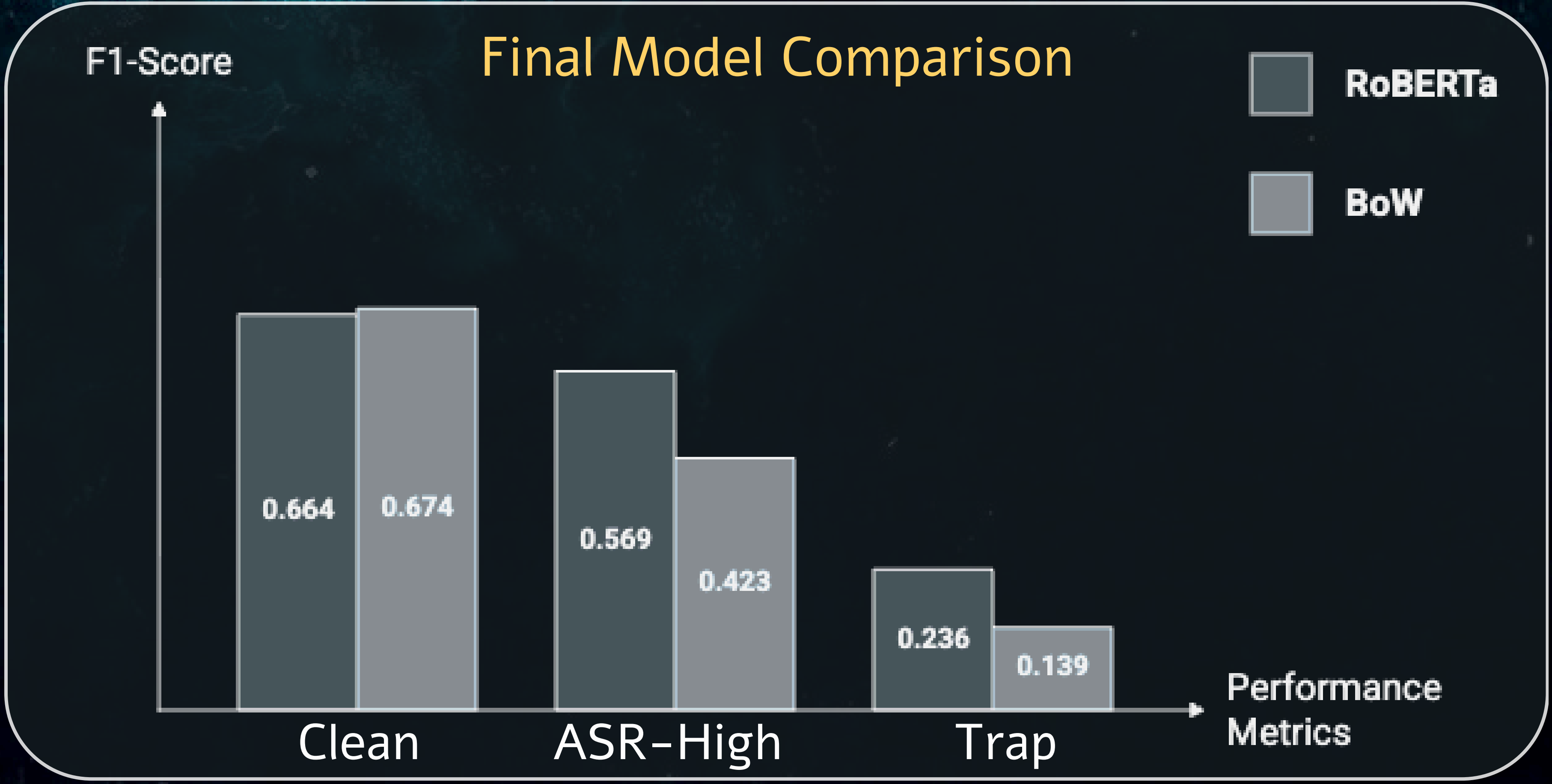
RoBERTa Final Performance:

F1 = 0.664 (clean) F1 = 0.569 (noisy)

- Transformers outperform traditional models
- Performance remains stable under noisy ASR conditions







Conclusion & Future Work

- The pipeline successfully implements an end-to-end pipeline that transforms noisy ASR transcripts into a structured JSON output, including vessel name, location, and nature of distress
- Improved classification of maritime messages (4 severity levels) under noisy ASR conditions
- Demonstrated improvement: RoBERTa is more robust than TF-IDF baselines to both adversarial phrasing and ASR transcription noise.

REMAINING CHALLENGES:

- Continued reliance on explicit codewords (e.g., MAYDAY)
- Further improvement of ASR robustness under high-noise conditions

IMPROVEMENT ACTIONS

- Increase Data Augmentation with Noisy ASR
- Reduce Dependence on Codewords
- Strengthen Contextual Learning
- Mitigate Error Propagation Across the Pipeline
- Re-evaluate Models Under Realistic Noise Conditions

ADDITIONAL EXPERIMENTS

- Fine-tune ASR for Maritime Vocabulary
- Add Adversarial Data Augmentation
- Scale and Diversify the Dataset
- Extend to Additional Languages and Acoustic Features